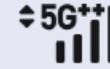


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SIMATS Online Programming Lab

PYTHON PROGRAMMING



KUMARAVEL D
192319019RD



Class Work(04.08.26)

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QUESTIONS

Q1

Student Database
5 marks



Q2

Menu Driven
5 marks



Q3

Exactly One Set
5 marks



Q4

Unique Words
5 marks



Q5

Unique Characters
5 marks



Q6

Pair Sum
5 marks



Q7

Duplicate Tuples
5 marks



Q8

Compare Lists
5 marks



Q9

Duplicate Elements
5 marks



Q10

First Repeated
5 marks



10/10 answered

Q1 Student Database

5 marks

Write a Python program to manage a student database using sets.

Sample Input/Output

Add Student

Input

Enter your choice: 1

Enter student name: Ravi

Output

Student added successfully.

Add Another Student

Input

Enter your choice: 1

Enter student name: Priya

Output

Student added successfully.

Display Students

Input

Enter your choice: 4

Output

Students: {'Ravi', 'Priya'}

Search Student

Input

Enter your choice: 3

Enter student name to search: Ravi

Output

Student found.

Remove Student

Input

Enter your choice: 2

Enter student name to remove: Ravi

Output

Student removed successfully.

Count Students

Input

Enter your choice: 5

Output

Total number of students: 1

Exit

Input

Enter your choice: 6

Output

Program terminated.

Your Code

```
1 students = set()
2
3 while True:
4     print("\nEnter your choice:")
5     print("1. Add Student")
6     print("2. Remove Student")
7     print("3. Search Student")
8     print("4. Display Students")
```

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QUESTIONS

Q1Student Database
5 marks**Q2**Menu Driven
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5 marks**Q4**Unique Words
5 marks**Q5**Unique Characters
5 marks**Q6**Pair Sum
5 marks**Q7**Duplicate Tuples
5 marks**Q8**Compare Lists
5 marks**Q9**Duplicate Elements
5 marks**Q10**First Repeated
5 marks

10/10 answered

Q2 Menu Driven

5 marks

Write a Python menu-driven program to perform set operations.

Sample Input

Enter the first set elements: 1 2 3 4

Enter the second set elements: 3 4 5 6

Menu

1. Union

2. Intersection

3. Difference (A - B)

4. Symmetric Difference

5. Exit

Enter your choice: 1

Sample Output

Union: {1, 2, 3, 4, 5, 6}

Sample Input

Enter your choice: 2

Sample Output

Intersection: {3, 4}

Sample Input

Enter your choice: 3

Sample Output

Difference: {1, 2}

Sample Input

Enter your choice: 4

Sample Output

Symmetric Difference: {1, 2, 5, 6}

Sample Input

Enter your choice: 5

Sample Output

Program terminated.

Your Code

```
1 A = set(map(int, input("Enter the first set elements: ").split()))
2 B = set(map(int, input("Enter the second set elements: ").split()))
3
4 while True:
5     print("\nMenu")
6     print("1. Union")
7     print("2. Intersection")
8     print("3. Difference (A - B)")
9     print("4. Symmetric Difference")
10    print("5. Exit")
11
12    choice = int(input("\nEnter your choice: "))
13
14    if choice == 1:
15        print("Union:", A.union(B))
16    elif choice == 2:
17        print("Intersection:", A.intersection(B))
18    elif choice == 3:
19        print("Difference:", A.difference(B))
20    elif choice == 4:
21        print("Symmetric Difference:", A.symmetric_difference(B))
22    elif choice == 5:
23        print("Program terminated.")
24        break
25    else:
26        print("Invalid choice! Please enter a number between 1 and 5.")
```

Input

1
2

Output

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5 marks**Q8**
Compare Lists
5 marks**Q9**
Duplicate Elements
5 marks**Q10**
First Repeated
5 marks

10/10 answered

Q3 Exactly One Set

5 marks

Write a Python program to find the elements that appear in exactly one of three sets.

Input:

A={1,2,3}

B={3,4,5}

C={5,6,7}

Output:

{1,2,4,6,7}

Your Code

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 C = {5, 6, 7}
4 result = (A | B | C) - ((A & B) | (A & C) | (B & C))
5 print('{' + ','.join(map(str, sorted(result))) + '}'')
```

Input

stdin input

Output

{1,2,4,6,7}

Visible Test Cases

Test Case 1



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Duplicate Tuples
5 marks

Q8
Compare Lists
5 marks

Q9
Duplicate Elements
5 marks

Q10
First Repeated
5 marks

10/10 answered

Q4 Unique Words

5 marks

Write a Python program to display the unique words in a sentence using a set.

Input:

python is easy python

{'python','is','easy'}

Your Code

```
1 sentence = input("Enter a sentence: ")
2 words = sentence.split()
3
4 unique_words = set(words)
5 print(unique_words)
```

Input

python is easy python

Output

Enter a sentence: {'python', 'easy', 'is'}

Visible Test Cases

Test Case 1

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5 marks

Q8
Compare Lists
5 marks

Q9
Duplicate Elements
5 marks

Q10
First Repeated
5 marks

10/10 answered

Q5 Unique Characters

5 marks

Write a Python program to count the number of unique characters in a string.

Input:

Programming

Output:

8

Your Code

```
1 s = input().strip()
2 print(len(set(s)))
```

Input

programming

Output

8

Visible Test Cases

Test Case 1



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Duplicate Tuples
5 marks

Q8

Compare Lists
5 marks

Q9

Duplicate Elements
5 marks

Q10

First Repeated
5 marks

10/10 answered

Q6 Pair Sum

5 marks

Write a Python program to find all pairs of elements whose sum equals a given number.

Input:

List:2 4 6 8 10

Target=12

ouput:

(4,8),(2,10)

Your Code

```
1 def pair_sum(arr, target):
2     arr.sort()
3     pairs = []
4     i, j = 0, len(arr) - 1
5
6     while i < j:
7         s = arr[i] + arr[j]
8
9         if s == target:
10             pairs.append((arr[i], arr[j]))
11             i += 1
12             j -= 1
13             while i < j and arr[i] == arr[i - 1]:
14                 i += 1
15             while i < j and arr[j] == arr[j + 1]:
16                 j -= 1
17         elif s < target:
18             i += 1
19         else:
20             j -= 1
21
22     return pairs
23
24
25 arr = [2, 4, 6, 8, 10]
26 target = 12
27
28 result = pair_sum(arr, target)
29 result.sort(key=lambda x: x[0], reverse=True)
30
31 print(", ".join(f'({a},{b})' for a, b in result))
```

Input

2 4 6 8 10

Output

(4,8),(2,10)

Visible Test Cases

Test Case 1



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Pair Sum
5 marks

Q7
Duplicate Tuples
5 marks

Q8
Compare Lists
5 marks

Q9
Duplicate Elements
5 marks

Q10
First Repeated
5 marks

10/10 answered

Q7 Duplicate Tuples

5 marks

Write a Python program to remove duplicate tuples from a list using a set.

Input:

[(1,2),(1,2),(3,4)]

Output:

[(1,2),(3,4)]

Your Code

```
1 lst = [(1,2),(1,2),(3,4)]  
2 result = list(set(lst))  
3 print(str(result).replace(" ", ""))
```

Input

stdin input

Output

```
[(1,2),(3,4)]
```

Visible Test Cases

Test Case 1



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Duplicate Tuples
5 marks**Q8**
Compare Lists
5 marks**Q9**
Duplicate Elements
5 marks**Q10**
First Repeated
5 marks

10/10 answered

Q8 Compare Lists

5 marks

Write a Python program to check whether two lists contain the same unique elements.

Input:

List1:1 2 2 3

List2:3 2 1

Output:

True

Your Code

```
1 list1 =[1,2,2,3]
2 list2 =[3,2,1]
3
4 if set(list1)==set(list2):
5     print(True)
6 else:
7     print(False)
```

Input

1 2 2 3
3 2 1

Output

True

Visible Test Cases

Test Case 1

0.00 / 5 marks

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5 marks**Q8**
Compare Lists
5 marks**Q9**
Duplicate Elements
5 marks**Q10**
First Repeated
5 marks

10/10 answered

Q9 Duplicate Elements

5 marks

Write a Python program to find all duplicate elements in a list using a set.

Input:

10 20 30 20 10

output:

{10,20}

Your Code

```
1 arr = list(map(int, input().split()))
2 seen = set()
3 dups = set()
4
5 for x in arr:
6     if x in seen:
7         dups.add(x)
8     else:
9         seen.add(x)
10
11 print("{* + ",".join(map(str, sorted(dups))) + "}")
```

Input

10 20 3 20 10

Output

{10,20}

Visible Test Cases

Test Case 1



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5 marks**Q7**
Duplicate Tuples
5 marks**Q8**
Compare Lists
5 marks**Q9**
Duplicate Elements
5 marks**Q10**
First Repeated
5 marks

10/10 answered

Q10 First Repeated

5 marks

Write a Python program to find the first repeated element in a list using a set.

Input:

10 20 30 20 40

Output:

20

Your Code

```
1 arr = list(map(int, input().split()))
2
3 seen = set()
4 for x in arr:
5     if x in seen:
6         print(x)
7         break
8     seen.add(x)
```

Input

10 20 30 20 40

Output

20

Visible Test Cases

Test Case 1



5.00 / 5 marks

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